

Dynamic Response of First Order Systems

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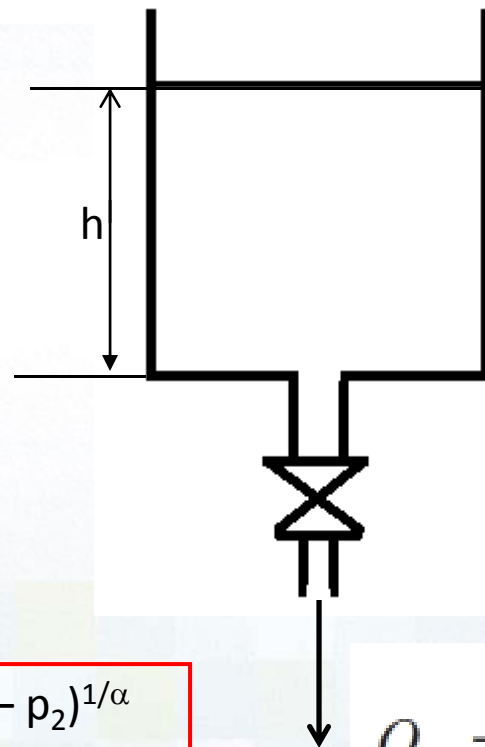
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This Lecture Contains

- A Few Examples of First Order Mechanical & Electrical Systems
- Response of a First Order system
- Unit Step Response
- Unit Ramp response
- Assignment Problem to Solve

Leaking Tank: A First Order System



$$\text{MFR} = 1/R (p_1 - p_2)^{1/\alpha}$$

$$\alpha = 1 \text{ for } \text{Re} < 1000$$

$$Q_{out} = \left(\frac{1}{R} \right) h$$

Considering Incompressible Fluid,
the Governing EOM :

$$d/dt (A h(t)) = -Q_{out} = (1/R)h(t)$$

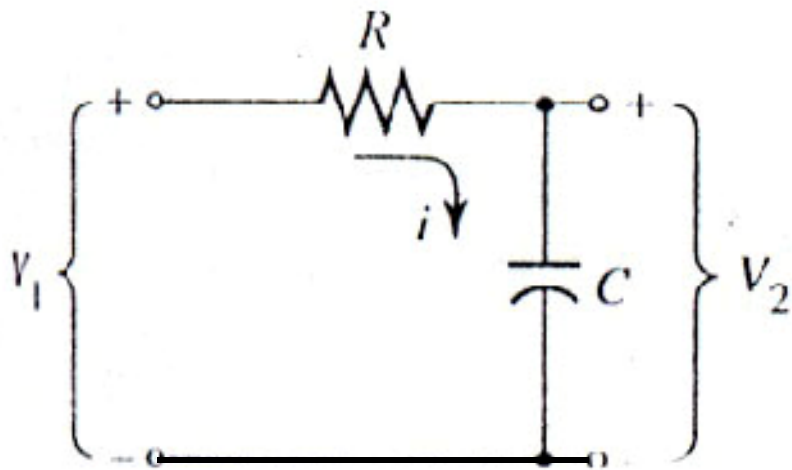
$$d/dt (h) = (1/AR) h$$

In standard form:

$$X=h, X_o = h_o, d/dt (X) = K X$$

A= Cross sectional
area of tank

A Low-Pass RC Filter

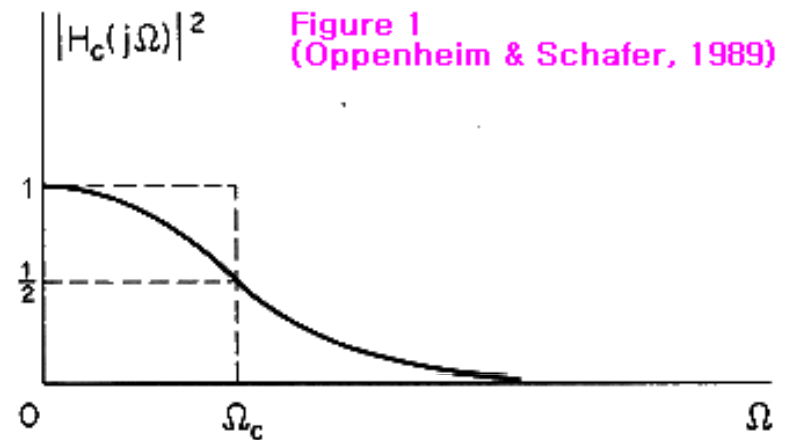


Low Pass RC Filter

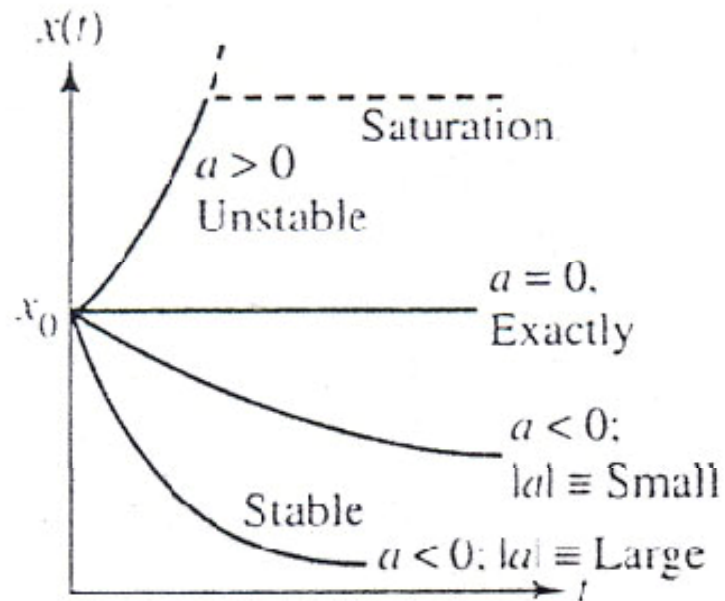
$$\frac{d}{dt}(V_2) = \frac{1}{RC}(V_1 - V_2)$$

$$V_1 = 0$$

$$\frac{d}{dt}(V_2) = \left(-\frac{1}{RC}\right)V_2$$



Free-Response of a First Order System



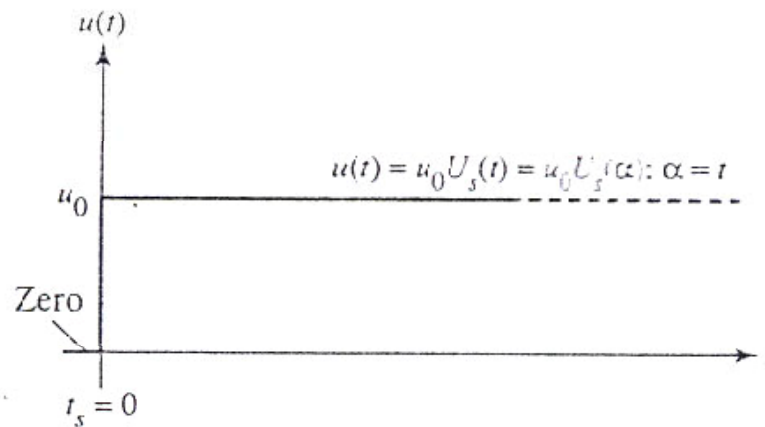
$$x(t) = e^{at} x_0$$

$a = 0$, Open circuit condition

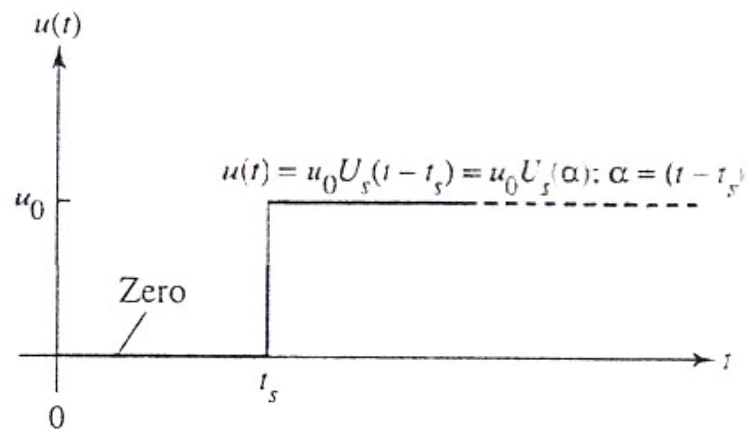
$T = 1/a$ time constant, time taken to reach $1/e$ of the initial value

Graph of e^{at} for ranges of a

Forced Excitation (Unit Step)

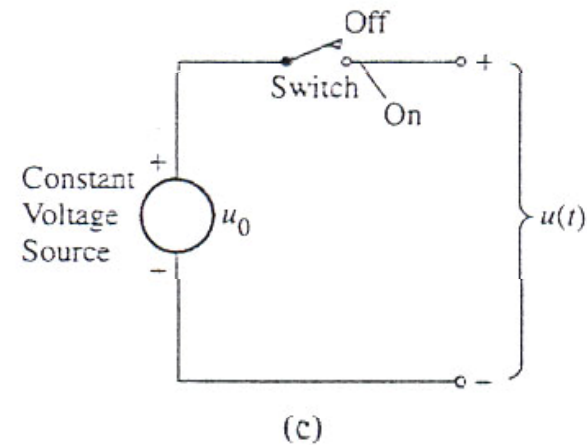


(a)



(b)

Step Input Function and a Switch as Function Generator

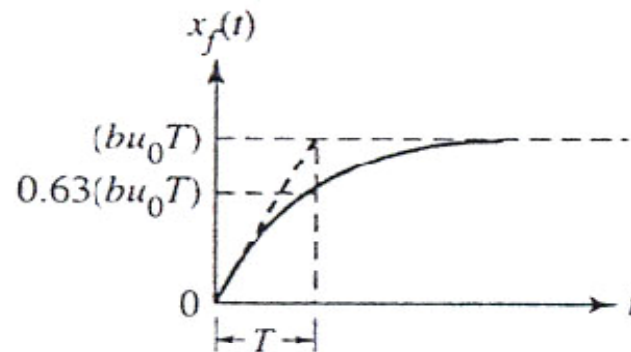


(c)

$$\frac{d}{dt} (x(t)) = a x(t) + b u(t)$$

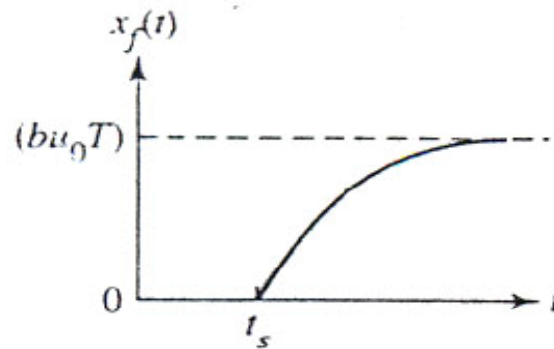
$$x_f(t) = \int_0^t e^{a(t-\tau)} b u(\tau) d\tau$$

Forced Response (Unit Step)



(a) $u(t) = u_0 U_s(t)$

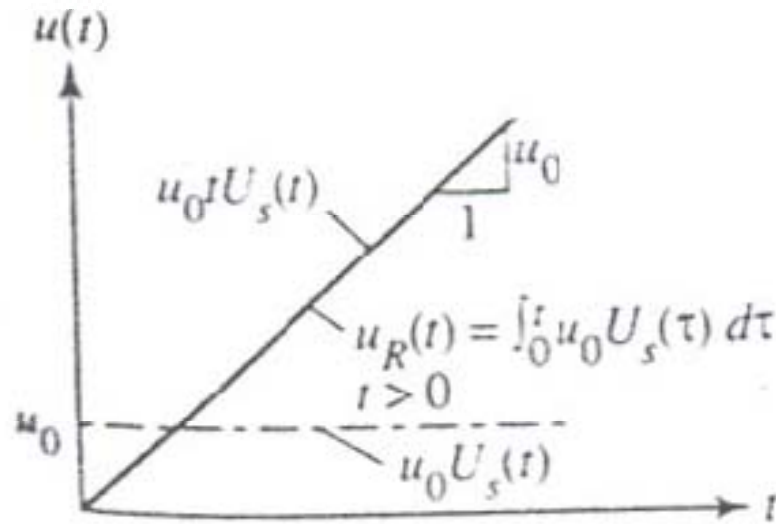
$$x_f(t) = (bu_0T)[1 - e^{-t/T}] U_s(t)$$



(b) $u(t) = u_0 U_s(t - t_s)$

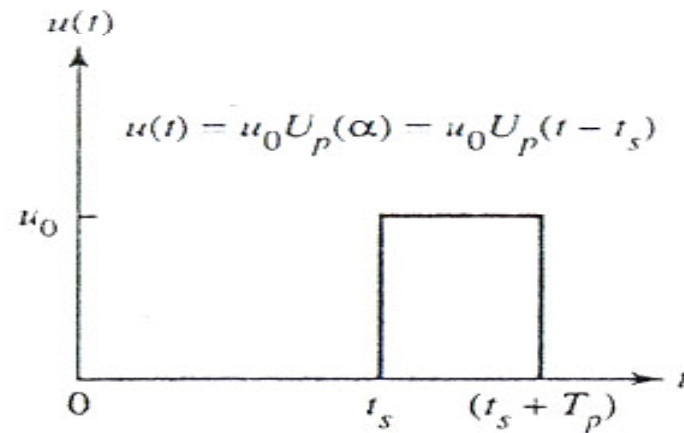
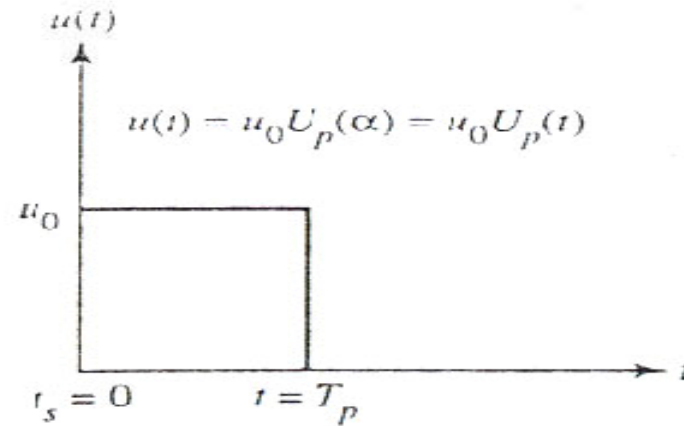
**Sketch of a Forced
Step Response**

Forced Excitation (Ramp Input)

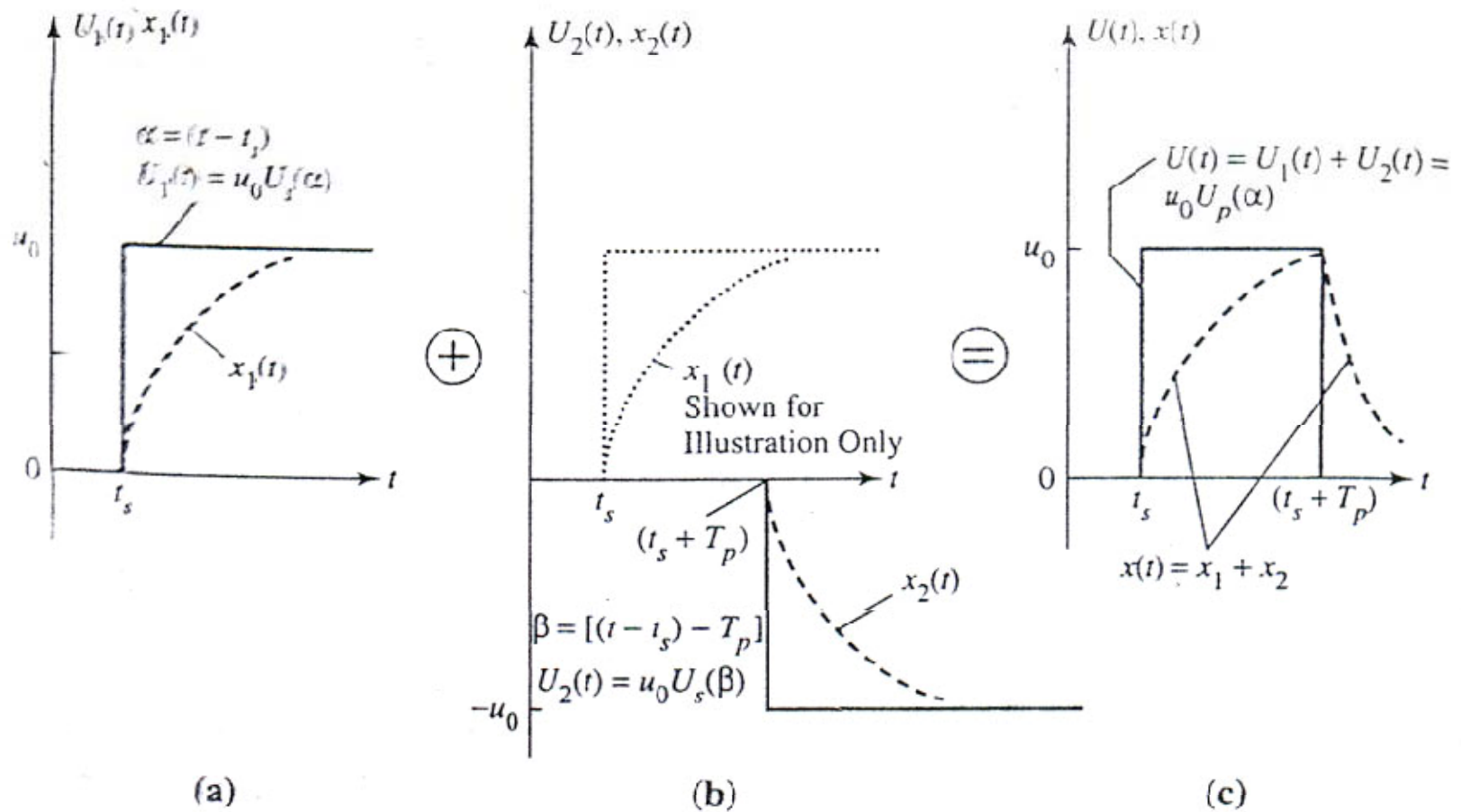


$$x_f(t) = bu_0 T^2 (e^{-t/T} + t/T - 1)$$

Forced Response (Pulse Input)



Pulse Response as sum of Step Responses



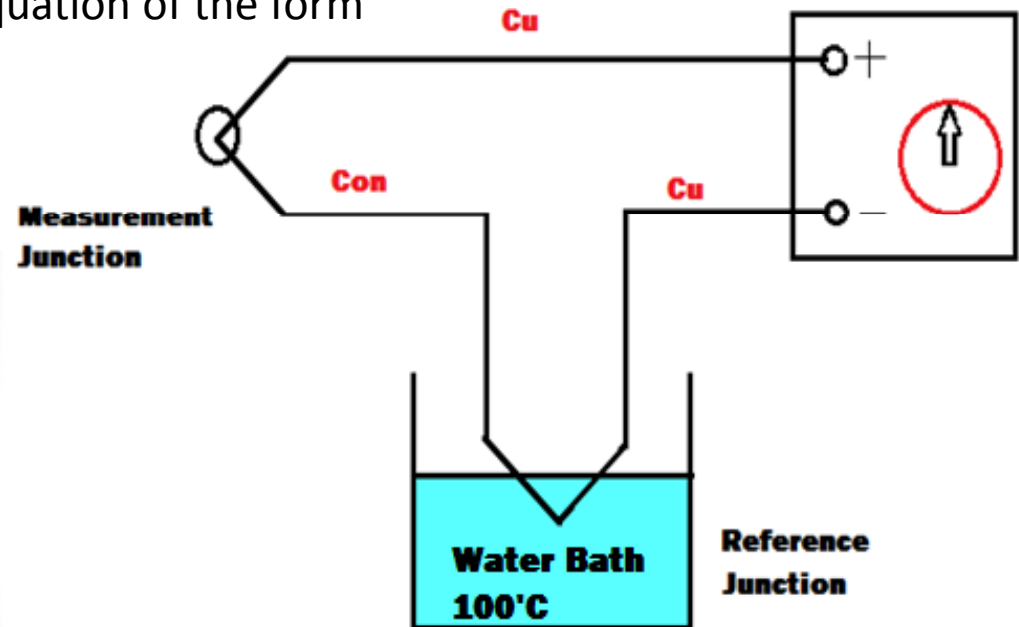
First Order Systems

A first order system has a differential equation of the form

$$\tau \frac{dy}{dt} + y = k r$$

$$Y(s) = G(s) R(s), G(s) = \left(\frac{1/\tau}{s + 1/\tau} \right)$$

$$y(t) = k \left(1 - e^{-t/\tau} \right)$$



Traditional Thermocouple Measurement

Example:

A thermocouple which has a transfer function linking its voltage output V and temperature input of T as

$$G(s) = \frac{30 \times 10^{-6}}{10s + 1} V/^{\circ}C$$

Determine the response of the system when it is suddenly immersed in a water bath at $100^{\circ}C$

The output as an ' s ' function is

$$V(s) = G(s) * \text{input}(s)$$

Sudden immersion of the thermometer gives a step input of size 100° C

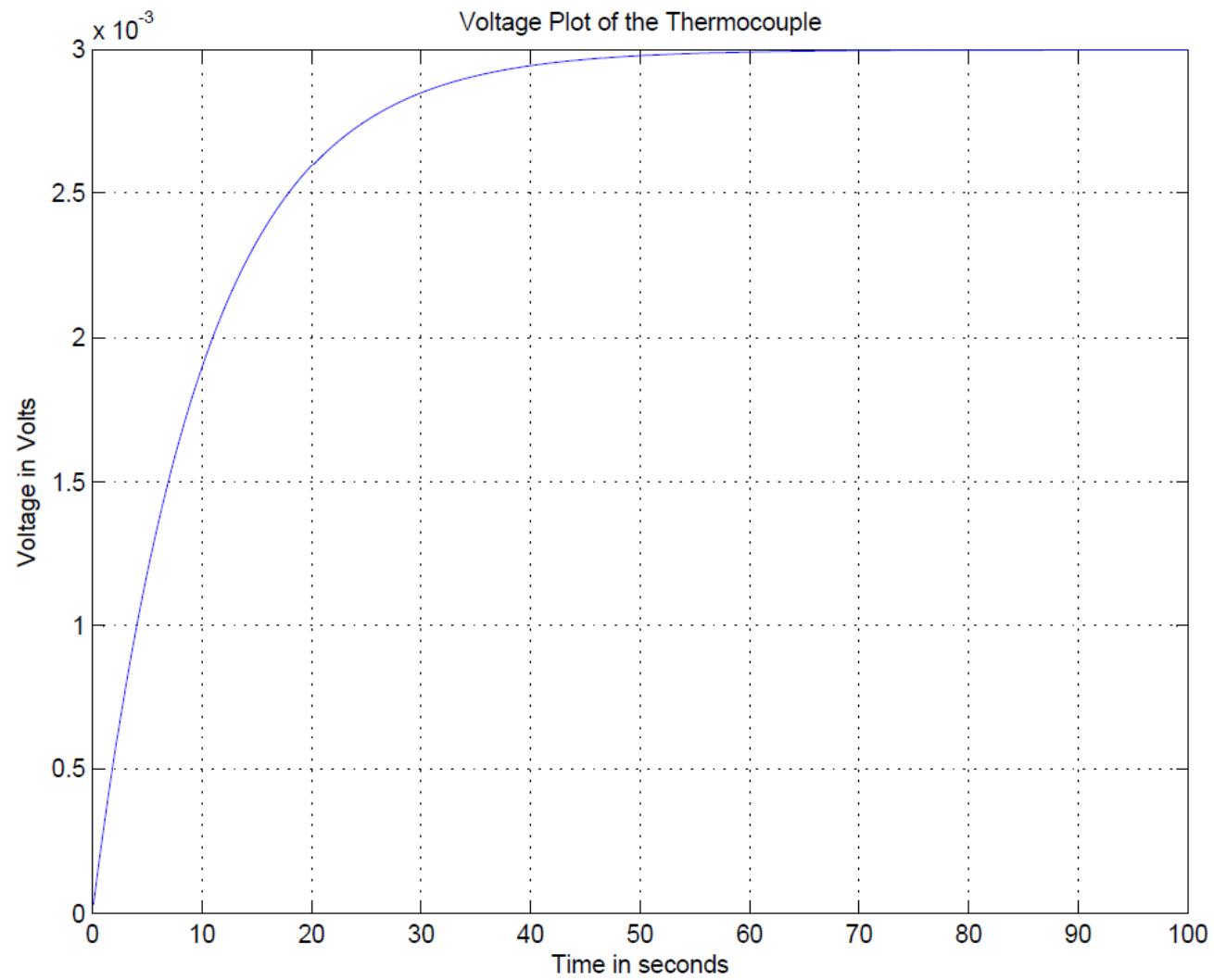
and so the input as an s function as $100/s$. Thus

$$V = \frac{30 \times 10^{-6}}{10s+1} \times \frac{100}{s} = \frac{30 \times 10^{-4}}{10s(s+0.1)} = \frac{30 \times 10^{-4} \times 0.1}{s(s+0.1)}$$

The fraction element of the form $a/s(s+a)$ and so the output as a function of time is :

$$V = 30 \times 10^{-4} (1 - e^{-0.1t})$$

The same is plotted in the following Figure. You may note the nature of the first order response. The Thermocouple took about a minute to reach close to the final value (about 3mv).



Assignment

Consider a first order system which can be simply modeled as a combination of a spring of stiffness k and damper with damping constant c connected in parallel. Find out the response of the system when it is subjected to an unit impulse excitation.

Special References for this lecture

- *System Dynamics for Engineering Students*: **Nicolae Lobontiu, Academic Publisher**
- *Feedback Control of Dynamic Systems*: **Frankline, Powell and Emami-Naeini, Pearson Publisher**
- *Control Systems Engineering*: **Norman S Nise, John Wiley & Sons**
- *Systems Dynamics and Response*: **S. Graham Kelly, Thomson Publisher**